

Dephasing Time T_2 in Quantum Systems

FYS5419

What is Dephasing?

Dephasing refers to the loss of phase coherence between quantum states due to interactions with the environment.

- Impacts off-diagonal elements of the density matrix.
- Leads to reduced visibility in interference experiments.
- A key limiting factor in quantum computing, sensing, and communication.

Density Matrix Perspective

For a qubit in state ρ :

$$\rho = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \text{ Dephasing primarily suppresses the coherence terms :}$$

$$\rho_{01}(t) = \rho_{01}(0)e^{-t/T_2}$$

Bloch Sphere Representation

- Dephasing corresponds to shrinking of the Bloch vector in the transverse plane.
- Longitudinal component (along z) is unaffected.

(Insert Bloch sphere illustration)

T_1 vs T_2 Times

- T_1 : Energy relaxation (population decay)
- T_2 : Phase relaxation (coherence decay)
- Always: $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}$

Sources of Dephasing

- Magnetic field fluctuations
- Charge noise
- Spin baths
- Phonons and lattice vibrations
- Classical low-frequency noise ($1/f$ noise)

Common noise spectral densities:

- White noise: constant spectral density
- $1/f$ noise: dominates at low frequencies
- Ohmic noise: linear in frequency
- Lorentzian noise: due to two-level fluctuators

Master Equation Formulation

Lindblad equation with dephasing: $=$
 $-i[H, \rho] + \gamma_{\phi}(\sigma_z \rho \sigma_z - \rho)$ Here : $T_2 = \frac{1}{2\gamma_{\phi}}$

Ramsey Experiment

Ramsey sequence measures free-induction decay:

- Prepare superposition
- Free evolution time t
- Final $\pi/2$ pulse

Signal: $S(t) = S_0 e^{-t/T_2} \cos(\Delta t)$

Spin Echo

Spin echo recovers coherence lost due to slow noise. $S(t) = S_0 e^{-t/T_2^{\text{echo}}}$ Typically: $T_2^{\text{echo}} > T_2$.

T_2 Time

- T_2 includes inhomogeneous broadening.
- Extracted from Ramsey without echo.
- Relation: $T_2^* < T_2 < T_2^{\text{echo}}$.

Typical T_2 Values: Superconducting Qubits

Examples:

- Transmons: $T_2 \sim 50\text{--}200 \mu\text{s}$
- Flux qubits: shorter due to flux noise

(Insert chart)

Typical T_2 Values: Trapped Ions

- T_2 times often exceed seconds
- Limited mainly by magnetic field drift

Typical T_2 Values: NV Centers in Diamond

- Room-temperature T_2 : 100–500 μs
- With isotopic purification: up to ms

Typical T_2 Values: Semiconductor Spin Qubits

- T_2^* often ns to μ s range
- Echo-improved T_2 : tens to hundreds of μ s

Mathematical Derivation: Free Induction Decay

Given Hamiltonian with fluctuating frequency: $H =$

$$\frac{1}{2}(\omega_0 + \delta\omega(t))\sigma_z$$

Phase: $\phi(t) = \int_0^t dt' \delta\omega(t')$ Coherence: $\langle e^{i\phi(t)} \rangle = e^{-t/T_2}$

Noise Correlation Functions

If noise is Gaussian: $T_2^{-1} = \int_0^\infty d\omega S(\omega)$ where $S(\omega)$ is the noise spectrum.

Experimental Signatures of Dephasing

- Linewidth broadening
- Amplitude decay in Ramsey/echo experiments
- Reduced gate fidelity

Role of Dephasing in Quantum Computing

- Limits coherence time window for gates
- Fundamental constraint on error correction
- Determines clock speed of logical operations

Mitigating Dephasing

- Dynamical decoupling sequences
- Material purification
- Cooling and shielding
- Optimal control pulses
- Noise spectroscopy and adaptive correction

Summary

- T_2 characterizes phase coherence
- Essential for quantum technologies
- Strong platform dependence
- Experimental control and theory needed for mitigation

Advanced Derivation of $\chi(t)$

Starting from the cumulant expansion for Gaussian noise:

$\chi(t) = \frac{1}{2} \int_0^t dt_1 \int_0^t dt_2 C(t_1 - t_2)$, where $C(\tau)$ is the noise autocorrelation.

Using its spectral density: $C(\tau) =$

$\frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega S(\omega) e^{-i\omega\tau}$. Combining yields the standard filter function representation

Filter Function Formalism

A control sequence defines a modulation function $y(t)$, leading to:

$$\chi(t) = \frac{1}{2\pi} \int_0^\infty d\omega S(\omega) \frac{|F(\omega)|^2}{\omega^2}, \text{ where } F(\omega) = \int_0^t dt' y(t') e^{i\omega t'}.$$

- Ramsey: $y(t) = 1$.
- Spin echo: $y(t) = 1$ for $t < \tau$, -1 afterwards.

Comparison of Control Sequences

- Ramsey: sensitive to low-frequency noise.
- Hahn Echo: cancels quasi-static noise.
- CPMG: multiple π pulses increase suppression bandwidth.

Include conceptual plots of $|F(\omega)|^2$.

Materials Origins of Dephasing in Superconducting Qubits

Sources:

- Two-level fluctuators in amorphous oxides.
- Flux noise from surface spins.
- Charge noise from dielectric loss.

Mitigations include surface cleaning, substrate engineering, and improved fabrication.

Dephasing in Trapped-Ion Systems

Dominant mechanisms:

- Magnetic field drifts.
- Laser phase fluctuations.
- Motional heating.

Use of sympathetic cooling and clock transitions minimizes noise.

NV Centers: Role of Spin Bath

The ^{13}C nuclear spin bath causes spectral diffusion.

- Echo improves T_2 by refocusing bath dynamics.
- Isotopic purification reduces nuclear spin density.

Semiconductor Spin Qubits: Charge Noise and Hyperfine Effects

Key mechanisms:

- Hyperfine fluctuations (GaAs).
- Charge noise modifying exchange energy.

Silicon qubits mitigate noise via isotopic purification and optimized gate stacks.

Extended Table: T_1 , T_2^* , and T_2 Across Platforms

Include table with:

- Superconducting qubits.
- Spin qubits.
- NV centers.
- Trapped ions.
- Neutral atoms.
- Donor spins.

Extracting Coherence Times: Practical Considerations

- Fit models: exponential, stretched exponential, Gaussian.
- Frequency detuning corrections.
- Accounting for SPAM errors.

Derivation: Noise Spectroscopy from CPMG

Peak sensitivity at $\omega_n \approx \frac{\pi n}{t}$, *allowing sampling of $S(\omega)$ across frequencies.*

Redfield tensor gives dephasing and relaxation rates:

$$\Gamma_{ij,kl} = \sum_{\alpha\beta} R^{\alpha\beta}{}_{ij,kl} S_{\alpha\beta}(\omega).$$

Used to model multi-level systems.

Relevance for Quantum Error Correction

- Coherence time sets maximum logical gate depth.
- Fault-tolerant thresholds require low dephasing rates.
- Strategies depend on dominant noise channel.

Documented Improvements in T_2

- 3D transmons: quality factors improved by orders of magnitude.
- Donor spins: T_2 up to 100 s.
- NV centers: millisecond-scale coherence.

Visualizing Dephasing on the Bloch Sphere

Show successive shrinking of transverse radius. Discuss connection to noise bandwidth.

Simulating Dephasing

- Classical noise generation (Ornstein–Uhlenbeck, $1/f$ noise).
- Solving stochastic Schrödinger equation.
- Averaging over many trajectories.

Open Quantum Systems: Overview

- Markovian vs non-Markovian dynamics.
- Master equations.
- Quantum trajectories.

Platform Comparison Summary

Evaluate:

- Dominant noise sources.
- Control techniques.
- Typical coherence times.
- Scalability considerations.

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